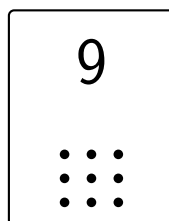
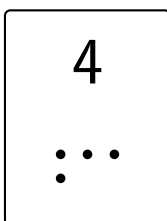
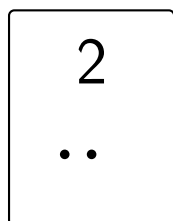


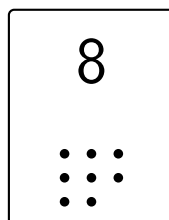
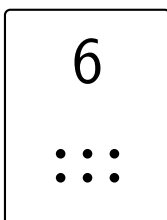
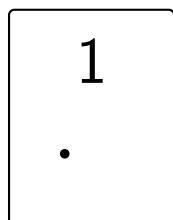
Each card has the same chance. Mix the bags separately. Draw once from each bag, then put both cards back. The bigger number wins.

Problem 1: Record every different pair of an A card and a B card, with its winner. Which deck wins more pairs?

A



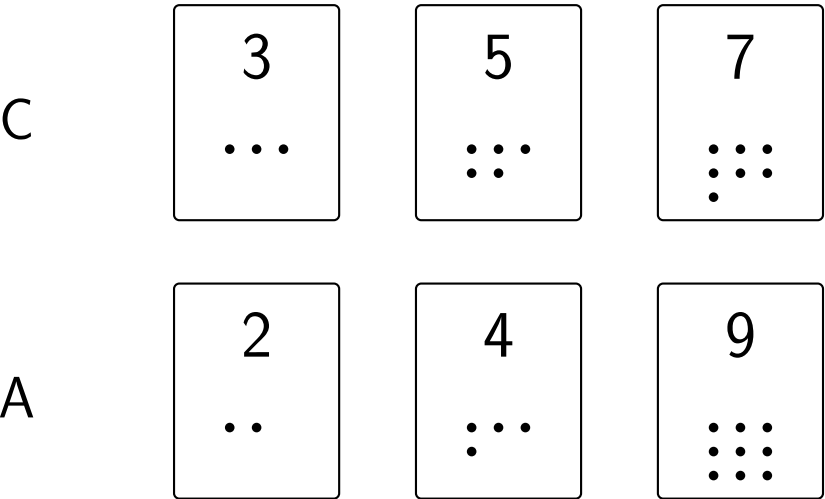
B



Problem 2: Record every different pair of a B card and a C card, with its winner. Which deck wins more pairs?

B	1 .	6 ⋮⋮	8 ⋮⋮⋮
C	3 ...	5 ⋮⋮	7 ⋮⋮⋮

Problem 3: Record every different pair of a C card and an A card, with its winner. Does any of A, B, and C win more pairs than both other decks?



Problem 4: For each of A, B, and C, choose a deck that wins more card pairs and play six rounds. Must your chosen deck win more rounds?

A	2 ..	4 :..	9 :::
B	1 .	6 :::	8 :::
C	3 ...	5 :::	7 :::

Problem 5: Try 3, 5, 7, and 9 in A's empty place. Which choices make A win more card pairs than B?

A

2

..

4

⋮

B

1

.

6

⋮

8

⋮

3

...

5

⋮

7

⋮

9

⋮

Problem 6: Swap one A card with one B card. Find every swap that makes B win more card pairs than A.

A	2 ..	4 :..	9 ⋮
B	1 .	6 ⋮	8 ⋮

Problem 7: Put these six cards into two decks of three so both decks win the same number of pairs. Find a way, or show why it cannot be done.

1 •	2 ••	3 •••	4 •••	5 •••	6 •••
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A

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B

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Problem 8: Can these six cards make three two-card decks where A wins more pairs than B, B more than C, and C more than A? Show a way, or show why it cannot be done.

1 •	2 ••	3 •••	4 •••	5 •••	6 •••
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A

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B

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C

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